

PROGRAM :

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.arima.model import ARIMA
np.random.seed(42)
date_range = pd.date_range(start='2023-01-01', end='2023-12-31', freq='D')
orders = 200 + 20*np.sin(np.linspace(0, 12*np.pi, len(date_range))) +
np.random.normal(0, 10, len(date_range))
data = pd.DataFrame({
    'date': date_range,
    'orders': orders
})
data.set_index('date', inplace=True)
data['orders'] = data['orders'].astype(int)
print("\nSample Data:\n", data.head())
plt.figure(figsize=(12,5))
plt.plot(data['orders'], label='Daily Orders')
plt.title('Food Delivery Demand Over Time')
plt.xlabel('Date')
plt.ylabel('Orders')
plt.legend()
plt.show()
decomposition = seasonal_decompose(data['orders'], model='additive',
period=30)
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decomposition.plot()
plt.show()
train = data[:'2023-10-31']
test = data['2023-11-01':]
model = ARIMA(train['orders'], order=(5,1,2))
model_fit = model.fit()
print("\nModel Summary:\n")
print(model_fit.summary())
forecast = model_fit.forecast(steps=len(test))
plt.figure(figsize=(12,5))
plt.plot(train.index, train['orders'], label='Train')
plt.plot(test.index, test['orders'], label='Actual')
plt.plot(test.index, forecast, label='Forecast')
plt.title('Forecast vs Actual')
plt.xlabel('Date')
plt.ylabel('Orders')
plt.legend()
plt.show()
future_forecast = model_fit.forecast(steps=30)
future_dates = pd.date_range(start=data.index[-1], periods=30, freq='D')
plt.figure(figsize=(12,5))
plt.plot(data.index, data['orders'], label='Historical')
plt.plot(future_dates, future_forecast, label='Future Forecast')
plt.title('Next 30 Days Demand Prediction')
plt.xlabel('Date')
plt.ylabel('Orders')
plt.legend()

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plt.show()

from sklearn.metrics import mean_absolute_error, mean_squared_error
mae = mean_absolute_error(test['orders'], forecast)
rmse = np.sqrt(mean_squared_error(test['orders'], forecast))
print("\nEvaluation Metrics:")
print("MAE:", mae)
print("RMSE:", rmse)
```